

Project Kavach

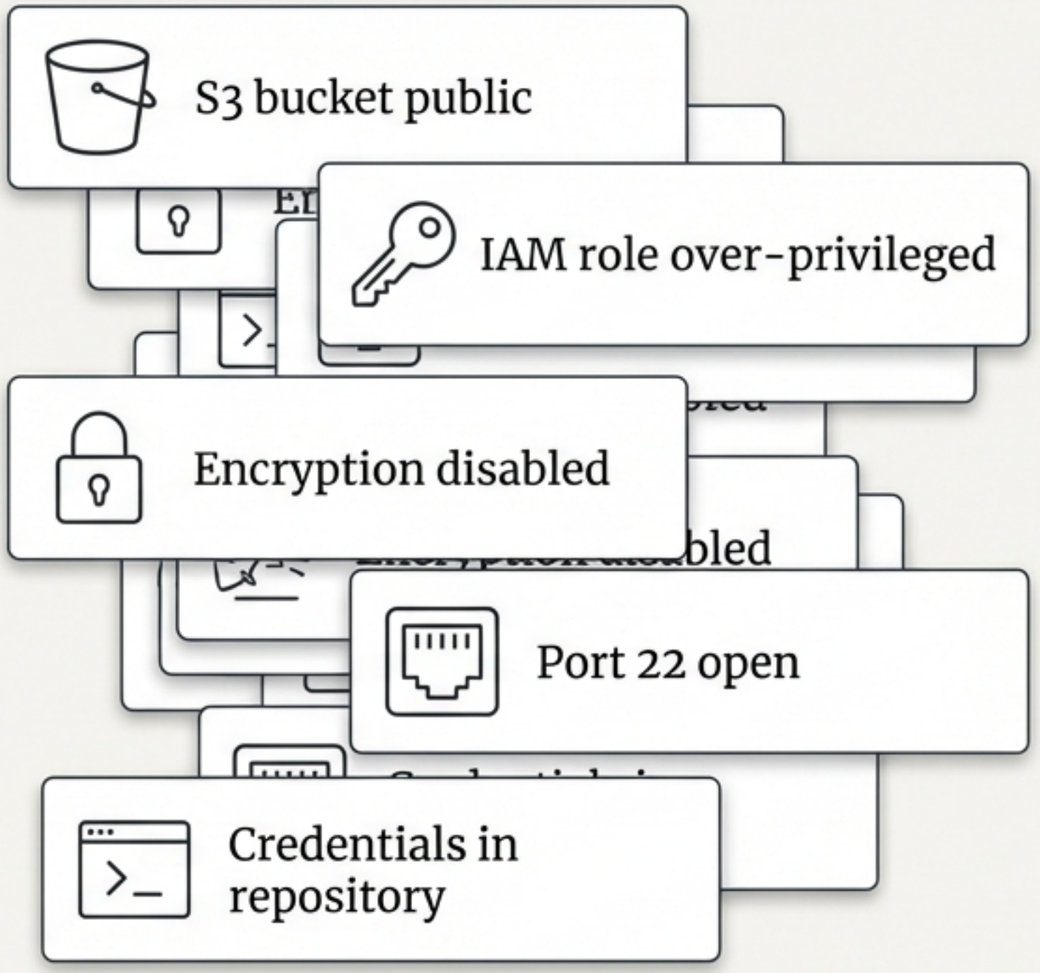
A Sovereign Zero-Trust Cloud Security Platform

From Misconfiguration Detection to Breach Simulation

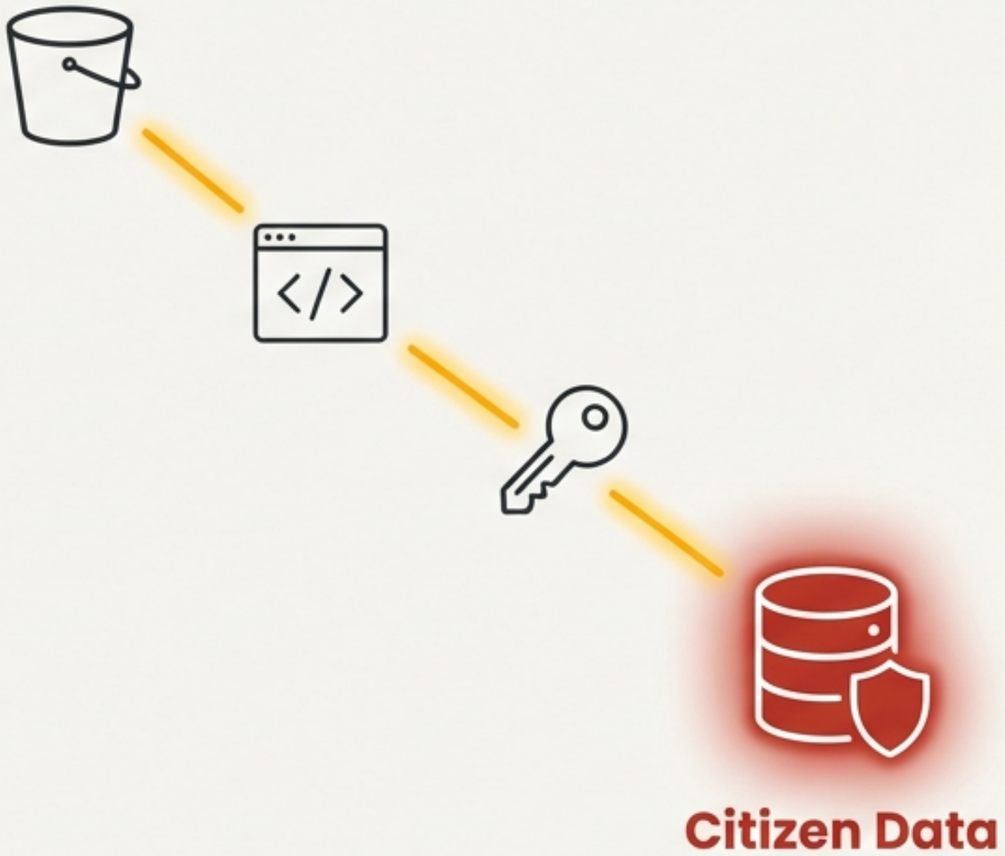
NotebookLM

Current Security Tools Give You a List. Attackers Execute a Plan.

The Checklist Trap



The Attacker's Reality



NotebookLM

An Attacker Sees a Path, Not a List

Misconfiguration Kill-Chain



> Your scanner says: 'S3 bucket is public ❌'.
> Kavach says: 'This public bucket leads to full account compromise in 3 steps ⚠️'.

NotebookLM

The Real Risk Isn't Just Severity, It's Speed



Time-to-Exploit (TTE)

The estimated time an attacker needs to exploit a misconfiguration once discovered.

Critical Risk Formula

If **TTE** (Time-to-Exploit) **<** **TTD** (Time-to-Detect) **=** **A CRITICAL BLIND SPOT**

"We prioritize what attackers can exploit fastest, not what looks severe on paper."

NotebookLM

Project Kavach: A Sovereign Shield for India's Digital Infrastructure

A sovereign, zero-trust cloud security posture management (CSPM) platform that detects, validates, and prioritizes cloud misconfigurations across public, private, and on-prem environments—using real breach attack paths, policy intent, and compliance-driven risk scoring.



Zero Trust by Design

Never trust, always verify, enforce least privilege.



Sovereign-Ready

Deployable on-prem or in air-gapped environments.



Compliance-First

Natively mapped to DPDP Act, CERT-In, MeitY, and RBI mandates.

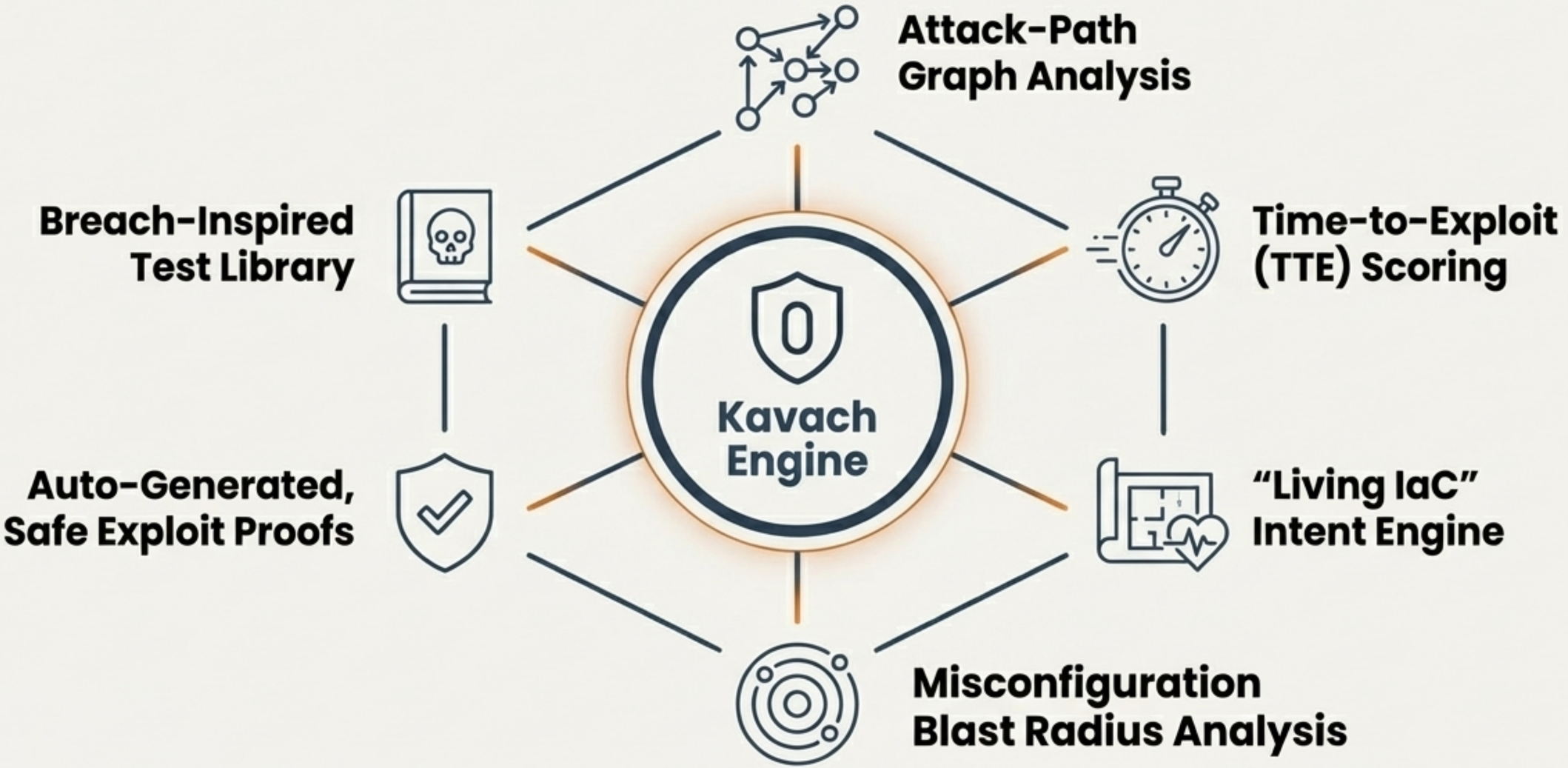


Least-Privilege Onboarding

We never require admin credentials.

NotebookLM

Six Layers of Attacker-Centric Intelligence

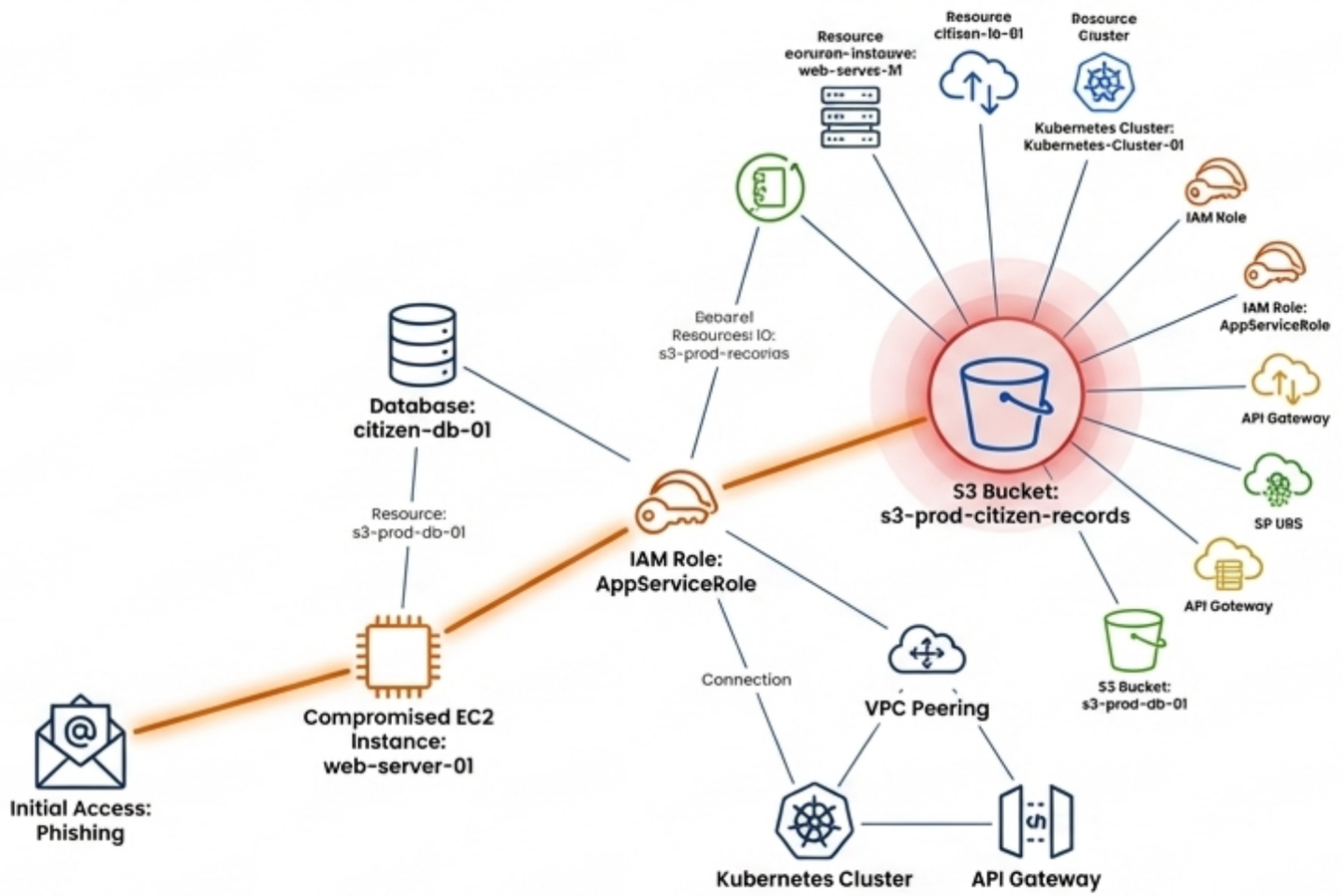


NotebookLM

Visualize the Full Attack Path and Quantify the Impact

Kavach

Kavach Attack Path Graph



Blast Radius Score

Resource: s3-prod-citizen-records

9.2 / 10



Impact on Services

- Citizen Aadhaar Verification API
- Public Grievance Portal
- Internal Analytics Dashboard

Data Sensitivity

PII | Financial Data | Sensitive

Regulatory Exposure

Digital Personal Data DPDPJ Act

CERT-In

NotebookLM

From Theoretical Risk to Actionable, Proven Findings

Detect Drift from Intent

Poppins SemiBold, Accent Blue (#2C3E50)



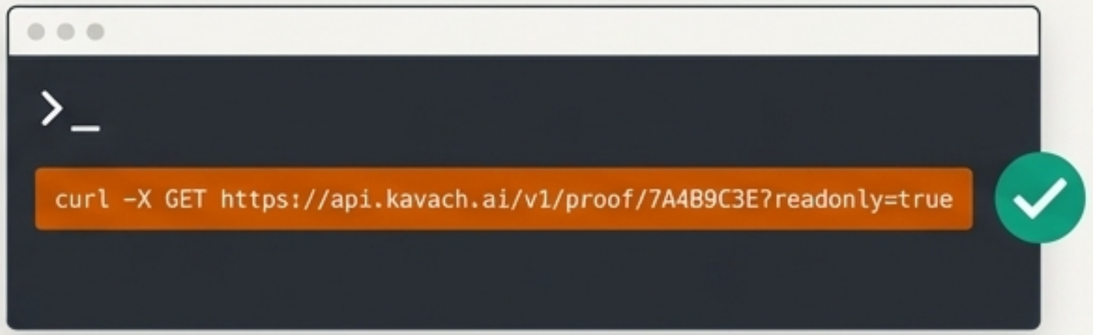
Concept Text: We detect deviations between Infrastructure-as-Code (IaC) templates and the live runtime environment.

Benefit Text: Identifies manual changes, uncodified hotfixes, and configuration drift.

"Misconfigs are often symptoms of broken workflows. We detect both."

Provide Safe Exploit Proofs

Poppins SemiBold, Accent Saffron (#D35400)



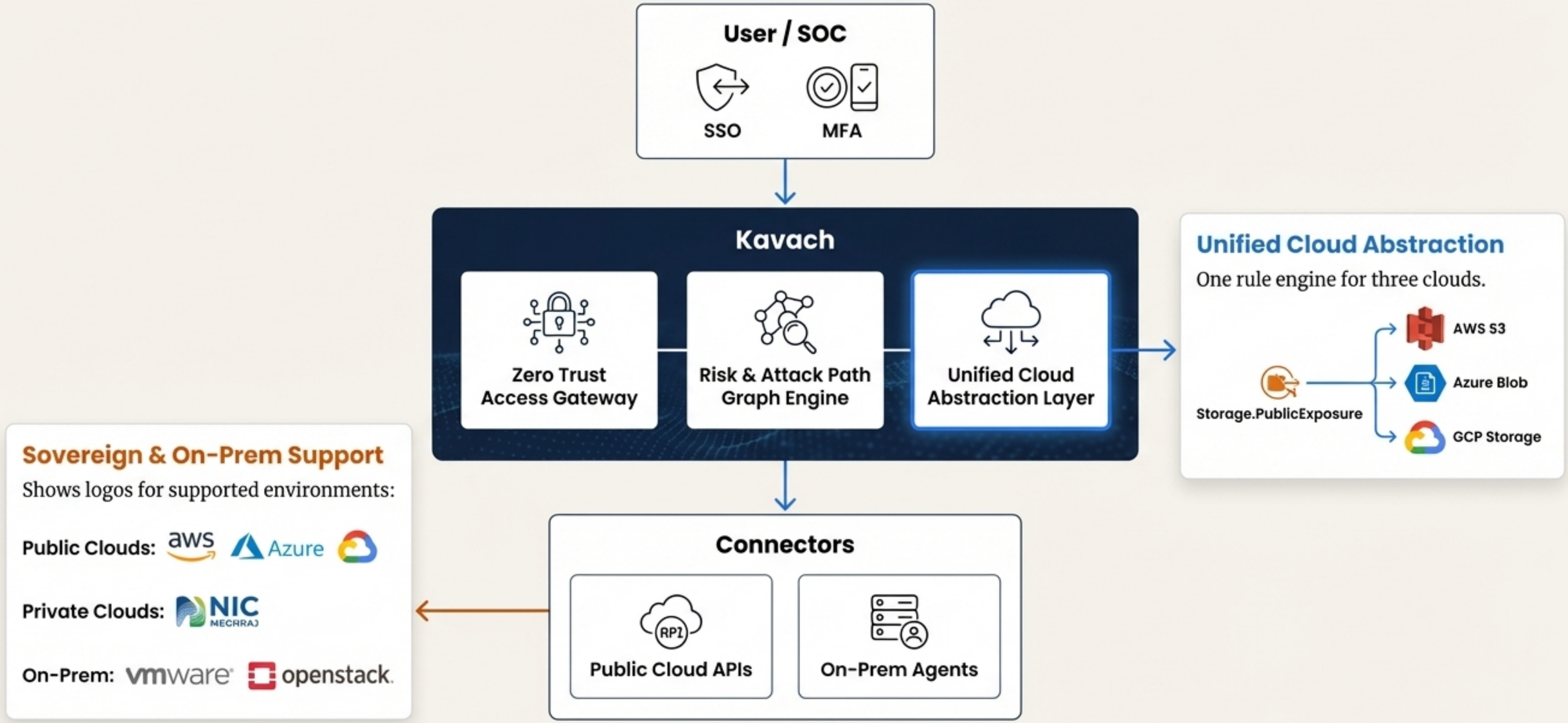
Concept Text: Merriweather Regular: For each critical finding, Kavach generates a safe, non-destructive, read-only proof.

Benefit Text: Eliminates false positives and provides engineers with definitive evidence of a vulnerability.

"Here is the exact API call an attacker would use."

NotebookLM

An Architecture Built for Hybrid, Sovereign, and Zero-Trust Operations



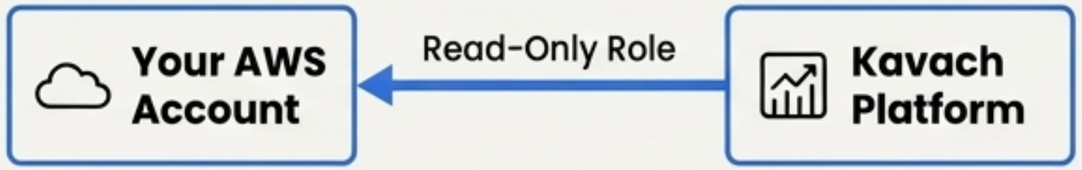
NotebookLM

Onboarding with Zero Trust from Day One

We Never Ask for Admin Credentials.

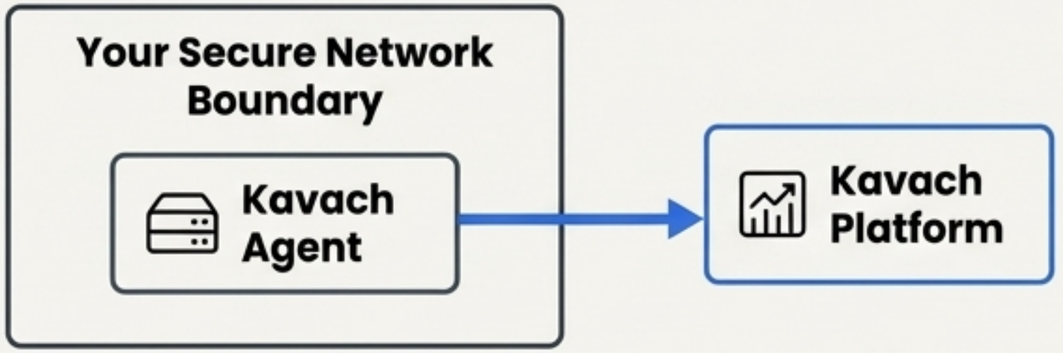
Public Cloud Onboarding (AWS Example)

- Cross-Account IAM Role using an External ID (to prevent confused-deputy attacks).
- Strictly `**Read-Only**` (`Describe*, `List*`).
- Fully auditable and revocable in seconds from your console.



On-Prem / Air-Gapped Onboarding

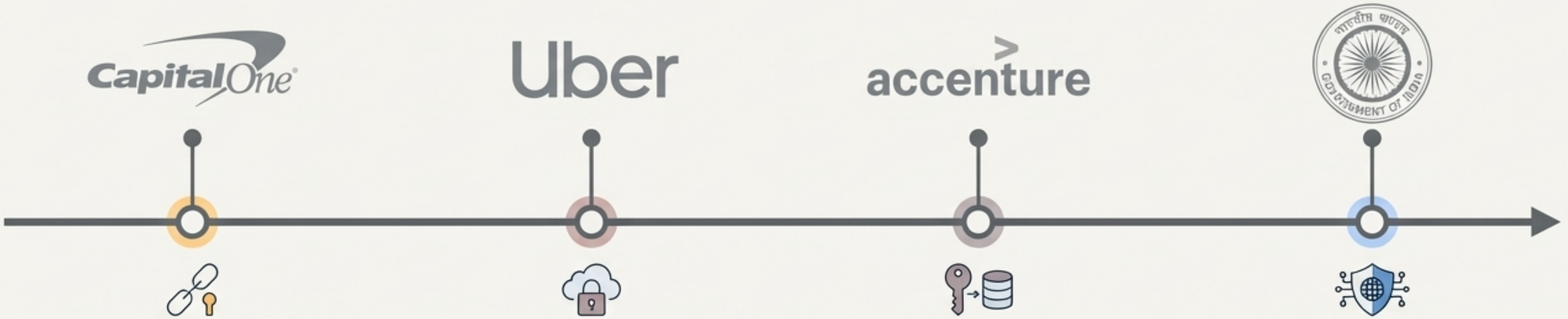
- Lightweight **scanner agent** runs inside your boundary.
- Pushes **metadata only** to the platform via a secure relay.
- No inbound internet access required. Ideal for secure defence networks.



NotebookLM

Our Test Library is Built from Real-World Breach Timelines

Our detection logic is not based on synthetic rules, but on the Tactics, Techniques, and Procedures (TTPs) of real-world attackers.



Every test case in our library maps to a verified, real-world attack.

NotebookLM

The Kavach Advantage: Translating Advanced Detection into Measurable Resilience

Key Performance Indicator (KPI)	How Kavach Improves
 **Reduced False Positives**	Attack-path correlation provides essential context, eliminating low-impact alerts.
 **Increased Detection Coverage**	The Unified Abstraction Layer provides consistent analysis across Public, Private, and On-Prem environments.
 **Reduced Detection Latency**	Near-real-time API polling and event-driven scans identify risks as they emerge.
 **Improved Compliance Readiness**	Built-in policy packs map findings directly to DPDP Act, CERT-In, and RBI controls.

NotebookLM

The Proposal: A Phased Path to Sovereign Cloud Resilience

Phase 1



Proof of Value (30 Days)

Scope: One critical application or cloud account.

Goal: Identify and validate top 5 critical attack paths.

Outcome: Demonstrate value and accuracy.

Phase 2



Expanded Deployment (90 Days)

Scope: Onboard additional departments and cloud environments.

Goal: Establish compliance baselines and integrate with SIEM/SOAR.

Outcome: Achieve broad visibility and operational integration.

Phase 3



Full Sovereignty (Ongoing)

Scope: Deploy on-prem or air-gapped instance.

Goal: Integrate with private clouds (e.g., NIC MeghRaj) and develop custom security policies.

Outcome: Achieve a fully sovereign, self-hosted security posture.

NotebookLM

Our Commitment: Securing India's Sovereign Digital Infrastructure

> "A **zero-trust, sovereign** cloud security platform that **detects misconfigurations** *not* in **isolation**, **but as real-world attack paths**—securely onboarded via least-privilege access and **deployable across** public, private, and **air-gapped government environments.**"

NotebookLM

Initiate Your Path to Proactive Cloud Resilience



Schedule a Live Technical Demo

See the Attack Path Graph and TTE scoring in action on a live system.



Initiate a 30-Day Proof of Value

Validate Kavach's detection capabilities within your own environment.



Request an Architectural Deep Dive

A dedicated session with our engineers to discuss on-prem, air-gapped, and hybrid deployment models.

Project Kavach Team | contact@project-kavach.gov.in | +91-11-2345-6789

NotebookLM